

EEB1303 – DIGITAL LOGIC CIRCUITS

IOT BASED ENERGY WASTAGE DETECTION AND NOTIFICATION SYSTEM

A PROJECT REPORT

Submitted by

ABISHEAK V	(927623BEE003)
BHARATHKUMAR R	(927623BEE014)
DHARNISH P S	(927623BEE022)
HARSINI R D	(927623BEE033)

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(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report “**IOT BASED ENERGY WASTAGE DETECTION AND NOTIFICATION SYSTEM**” is the Bonafide work of **ABISHEAK V (927623BEE003), BHARATHKUMAR R(927623BEE014) DHARNISH P S (927623BEE022), HARSINI R D (927623BEE033)** who carried out the project work during the academic year 2025-2026 under my supervision.

SIGNATURE

Dr. J. UMA M.E., Ph.D.

HEAD OF THE DEPARTMENT

Department of Electrical and
Electronics Engineering,
M.Kumarasamy College of Engineering,
Thalavapalayam, Karur-639113

SIGNATURE

Dr. S. SATHISH KUMAR M.E., Ph.D.

SUPERVISOR

Associate Professor,
Department of Electrical and
Electronics Engineering,
M.Kumarasamy College of Engineering,
Thalavapalayam, Karur-639113.

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EXTERNAL EXAMINER

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PEO3: Graduates will be a successful entrepreneur in creating jobs related to Electrical and Electronics Engineering /allied disciplines.

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- **PO1 Engineering knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop solutions to complex engineering problems.
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- **PO4 Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of

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- **PO5 Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
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- **PO11 Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

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- **PSO2:** Apply relevant models, resources and emerging tools and techniques to provide solutions to power and energy related issues & challenges.
- **PSO3:** Design, Develop and implement methods and concepts to facilitate solutions for electrical and electronics engineering-related real-world problems.

Abstract (Key words)	POs Mapping
This work presents a low-cost IoT-based system that detects energy wastage by monitoring human presence and electrical load using an ESP32, ultrasonic sensor, and ACS712 current sensor. It automates power control and sends instant Telegram alerts for efficient energy management.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11 PSO1, PSO2, PSO3

SDG Goal		Remarks
SDG 7	Affordable and Clean Energy	This project effectively minimizes energy wastage by automatically controlling electrical loads using real-time human detection. It contributes to improved energy efficiency and sustainability

ABSTRACT

Energy wastage has become a growing concern in modern infrastructures such as homes, hostels, institutions, and corporate buildings, where electrical appliances often remain active even in the absence of human presence. This unnecessary consumption not only increases electricity bills but also contributes to environmental issues such as excessive carbon emissions and resource depletion. With the increasing adoption of automation and smart systems, there is a strong need for an intelligent, affordable, and reliable solution that can continuously monitor usage conditions and prevent avoidable power loss. This project introduces an IoT-based Energy Wastage Detection and Notification System that focuses on enhancing energy efficiency through smart monitoring and timely user awareness. The system integrates human presence detection with real-time load measurement to understand when power is being consumed without occupancy. By correlating these two critical factors, the model identifies potential wastage scenarios and ensures that the user is immediately informed. This approach allows for proactive energy management. The proposed model is designed to be low-cost, scalable, and suitable for various environments ranging from small rooms to large institutional setups. By utilizing IoT connectivity, the system extends its functionality beyond basic detection and provides instantaneous alerts to the user's device, promoting quick corrective action. Overall, this project demonstrates how intelligent sensing, IoT communication, and automated decision-making can significantly improve energy conservation. It highlights an effective and practical solution that contributes to cost savings, user convenience, and a more sustainable future by ensuring that electrical energy is consumed only when truly required.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
ESP32	Espressif Systems 32-bit Microcontroller
HC-SR04	Ultrasonic Distance Measurement Sensor
ACS712	Allergo Current Sensors 712 Series
IoT	Internet of Things
ADC	Analog to Digital Converter
API	Application Programming Interface
RLM	Real-Time Load Monitoring
UPD	User Presence Detection
TAN	Telegram Alert Notification

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Energy conservation has become a vital concern in modern infrastructures, as a considerable amount of electricity is wasted when appliances such as lights and fans are left operating in the absence of people. With the introduction of IoT technologies, it is now possible to monitor, control, and optimize energy usage automatically. The proposed system effectively combines presence detection with load monitoring to identify real-time energy wastage and respond accordingly [8]. During implementation and testing, the system showed reliable performance in recognizing when a space was occupied and when it was empty. Whenever no presence was detected, the load remained OFF, ensuring that no unnecessary power was consumed. When presence was detected, the system immediately activated the load, demonstrating fast response and accurate decision-making [2].

The monitoring aspect also performed consistently by clearly reflecting changes in load consumption during ON and OFF states, confirming that the system was functioning correctly at every moment. Another key observation was the consistent communication capability of the system, which delivered timely updates to the user regarding presence detection, load status, and overall system activity [4]. This ensured that the user always remained informed, even from a remote location. The combination of accurate detection, efficient load control, and real-time notifications highlights the practical value of the system in reducing unnecessary energy usage. Overall, the testing results confirm that the project achieves its intended objectives by providing a dependable, automated, and user-friendly method for managing energy effectively in everyday environments.

1.2 PROBLEM STATEMENT

Electrical energy wastage due to unattended appliances has become a critical problem in both domestic and commercial environments. In many situations, lights, fans, and other electrical loads remain switched ON even when no human presence is detected, primarily because users either forget to turn them OFF or lack proper awareness regarding energy conservation [7]. This issue is widespread in classrooms, hostels, offices, shopping complexes, workshops, laboratories, and even residential settings where appliances operate for longer durations than necessary. Such continuous operation of electrical equipment contributes significantly to increased electricity bills, reduced appliance lifespan, and unnecessary environmental pollution through excessive energy generation demands [3] .

While awareness programs and manual monitoring may reduce wastage to some extent, they are neither fully reliable nor sustainable, as human errors cannot be eliminated completely. Therefore, there is a pressing need for a smart and automated system that can detect human presence accurately and regulate electrical loads without depending on manual intervention [5]. An effective solution must be capable of identifying whether a person is present in a particular area and must have the ability to immediately switch OFF appliances when the space becomes vacant. Additionally, the system should be able to monitor actual current consumption to provide insights into energy usage patterns.

Modern IoT technologies offer promising solutions by enabling smart devices to communicate, process data, and operate remotely with minimal human involvement [1]. Leveraging these capabilities, this project aims to build an intelligent human-presence-based load control system using an Ultrasonic Sensor, ESP32 microcontroller, Relay Module, and ACS712 current sensor. The ultrasonic sensor serves as the primary

detection mechanism by measuring the distance to an object or person in front of it, and based on this information, the ESP32 determines whether the electrical load should be turned ON or OFF. The ACS712 sensor measures current flow to confirm the operational status of the load and detect anomalies if present. To enhance usability and control, the system integrates Telegram communication to send real-time notifications to the user regarding human presence, power status, and load conditions. This enables remote monitoring and ensures that the user is aware of energy usage even when they are not physically present at the location [6]. Thus, the problem addressed by this project is the persistent and significant issue of energy wastage due to ineffective human-dependent control systems, and the solution proposed is a fully automated smart energy control system designed to reduce wastage, improve efficiency, and support sustainable energy practices.

1.3 OBJECTIVE OVERVIEW

The central objective of this project is to develop a fully automated, intelligent, and efficient energy management system capable of detecting human presence and controlling electrical loads accordingly. The project aims to design a system that functions independently without requiring manual switching, thereby reducing energy wastage and improving overall operational efficiency. By integrating multiple sensors and IoT functionalities, the system seeks to offer a comprehensive solution that not only automates load control but also monitors real-time current consumption and provides timely communication to users[8]. A key objective is to utilize the Ultrasonic Sensor to measure the distance of objects or humans within a specific range and interpret these measurements to determine whether the room is occupied or empty. This allows the system to activate the relay and switch ON the load only when required.

Another major objective is the use of the ACS712 current sensor, which serves as a reliable tool for detecting the actual power consumption and ensuring that the system operates safely and consistently. The ESP32 microcontroller plays a crucial role in processing sensor inputs, executing decision-making algorithms, and controlling the relay

output with precision. In addition to automation and sensing, the project aims to achieve seamless IoT-based communication by implementing Telegram Bot notifications [4]. This feature ensures that the user receives instant updates on load status, presence detection, and energy usage, enabling effective remote monitoring. Furthermore, the system aims to demonstrate the potential of IoT for sustainable energy management by reducing unnecessary appliance operation, minimizing human involvement, and promoting environmental consciousness [2]. The overarching objective is not merely to automate a single load but to establish a scalable, adaptable foundation for smart building technologies where multiple sensors and loads can be integrated to create a complete energy-efficient ecosystem [7]. Through this project, the objective is to showcase how embedded systems and IoT technologies can be practically applied to solve real-world problems and contribute to long-term energy sustainability.

1.4 SCOPE OF THE PROJECT

The scope of the IoT-based Energy Wastage Detection and Notification System is both broad and adaptable, extending from small residential settings to large-scale commercial and institutional environments. In residential homes, the system can be deployed in individual rooms to monitor and control energy-consuming devices, thereby ensuring that power is only used when needed and significantly reducing electricity bills. In larger commercial buildings, such as offices, schools, or hospitals, the system can be scaled up to cover multiple rooms and floors, providing comprehensive energy management across the entire facility [3]. The system's modular design means it can be easily customized and expanded, allowing for the addition of more sensors or integration with other smart home or building automation systems. The project also emphasizes ease of implementation and maintenance, ensuring that users can adopt the system without requiring extensive technical expertise [9]. Moreover, the project's scope includes future enhancements, such as data logging to cloud services like Google Sheets or Firebase, and the integration of additional alert mechanisms, such as LCD displays or buzzers. Overall, the scope of the project is designed to ensure flexibility, scalability, and long-term sustainability in energy management.

1.5 ADVANTAGE

The proposed human-presence-based energy management system offers numerous advantages that significantly enhance its practicality, efficiency, and usability in real-world environments. One of the most important advantages is the substantial reduction in electricity wastage, as the system ensures that loads operate only when human presence is detected and remain OFF when the area is unoccupied. This automatic switching eliminates dependency on human behaviour and minimizes the chances of appliances being left ON accidentally [1]. Another major advantage is the improvement in user convenience, as the automation reduces the need for manual switching and allows users to focus on their activities without worrying about monitoring appliances. The continuous current monitoring provided by the ACS712 sensor offers safety, efficiency, and real-time insights into energy consumption, which helps detect abnormalities and maintain appliance health.

The integration of IoT technology through Telegram notifications adds a modern communication layer that keeps users informed about load status at all times, enabling remote monitoring even when they are away from the location. This ensures better control and awareness, making the system suitable for both residential and institutional environments. The project is cost-effective because it uses readily available, low-cost sensors and components while delivering high functionality [10]. Its modular nature allows for easy expansion by adding more sensors, loads, or IoT features. In addition, the automated control reduces the wear and tear on electrical appliances, potentially increasing their lifespan by ensuring they operate only when needed. From an environmental perspective, the system contributes to sustainable energy usage by promoting responsible consumption and reducing unnecessary operational hours. Its scalability makes it useful not only for small rooms but also for large buildings with multiple zones [5]. Overall, the system provides a modern, efficient, and intelligent solution for improving energy management while enhancing user comfort, reducing costs, and supporting eco-friendly practices.

CHAPTER 2

LITERATURE REVIEW

PAPER TITLE: LITERATURE REVIEW ON IOT-BASED SOLAR TRACKING SYSTEMS USING LDR AND CLOUD MONITORING,

AUTHOR: E. Uhlmann et al,

YEAR: 2024,

DESCRIPTION:

This paper presents an advanced IoT-based energy monitoring framework designed specifically for industrial production environments where energy wastage frequently occurs due to idle equipment, inefficient load scheduling, and lack of real-time visibility of power usage. The authors implement a distributed sensor network equipped with current sensors, voltage sensors, and machine-status detectors to continuously record consumption parameters across multiple industrial machines[. The data is processed using an IoT gateway that communicates with cloud platforms for centralized analytics. The study emphasizes that conventional energy audits conducted manually or periodically are insufficient for identifying real-time inefficiencies. With IoT integration, the system provides continuous monitoring, anomaly detection, and automated alerts for abnormal loads. Real-time dashboards allow production managers to track machine-wise consumption, peak loads, and idle-time losses. The results demonstrate significant reductions in operational waste and improved energy planning, establishing IoT as a transformative tool for industrial power optimization.

PAPER TITLE: IoT Enabled Intelligent Energy Management System,

AUTHOR: C. K. Rao et al.,

JOURNAL: Sustainable Computing,

YEAR: 2024.

DESCRIPTION:

This research introduces an intelligent energy management system designed to optimize power consumption across residential, commercial, and institutional buildings using IoT technologies. The system integrates smart sensors, microcontrollers, and embedded decision-making algorithms capable of adjusting device operation based on occupancy, load condition, and environmental parameters. The authors highlight the limitations of older systems that rely on timers or manual switching, which often result in unnoticed wastage. By contrast, the proposed IoT-enabled system monitors appliances continuously and makes autonomous decisions such as turning loads ON/OFF, sending notifications, and generating usage reports. Cloud storage is used for long-term analysis, enabling predictive maintenance and peak-load forecasting. Experimental results show considerable savings in electricity usage, particularly in areas with fluctuating occupancy patterns such as offices, schools, and labs. The study demonstrates how IoT intelligence, combined with automated controls, can achieve sustainable energy consumption and support future smart-building applications.

PAPER TITLE: IoT-Enabled Occupancy-Based Energy Wastage Detection System Using ESP32

AUTHOR: R. Sharma et al.,

JOURNAL: IEEE International Conference on Smart Automation and Energy,

YEAR: 2024.

DESCRIPTION:

This study presents an IoT-enabled energy wastage detection framework that intelligently correlates occupancy data with electrical load consumption to identify unnecessary power usage in indoor environments such as classrooms, offices, and residential buildings. The system employs an ESP32 microcontroller due to its built-in Wi-Fi, low power consumption, and capacity to interface with multiple sensors. An ultrasonic sensor is used to detect human presence by precisely measuring distance, avoiding the false triggers commonly seen in PIR-based human detection. Simultaneously, a Hall-effect ACS712

current sensor measures real-time current flow through loads such as lights, fans, and appliances. The system identifies wastage by detecting conditions where current is flowing but no human presence is detected, thereby automating the recognition of idle loads. Sensor data is uploaded to a cloud or mobile interface, enabling remote monitoring and automated decision-making. Alerts are sent instantly through IoT communication platforms to warn users of detected wastage, allowing quick corrective action. Experimental results demonstrate a significant reduction in power wastage during unoccupied periods, especially in institutions and workplaces where users often forget to switch off appliances. The research concludes that IoT-based occupancy–power correlation is a highly efficient, low-cost, and scalable solution for sustainable energy management in modern buildings.

PAPER TITLE: Real-Time IoT-Based Power Monitoring with Intelligent Alert Notification Using ACS712 Sensor

AUTHOR: M. Kaleem et al.,

JOURNAL: International Journal of Smart Energy Systems (IJSES)

YEAR: 2025

DESCRIPTION:

This research proposes an intelligent IoT-based real-time power monitoring and alert-generation system aimed at identifying abnormal or unnecessary energy consumption. The model uses the ACS712 current sensor to capture instantaneous electrical load characteristics and transmit the data to an ESP32 microcontroller for processing. The system calculates real-time current, voltage, and power consumption values, enabling precise tracking of load behaviour. A notable contribution of this study is the integration of instant alerting mechanisms using lightweight IoT communication tools such as Telegram Bot API and MQTT notification systems. These alerts notify users immediately when energy wastage, unattended loads, or abnormal consumption patterns are detected, enabling rapid corrective measures. The authors emphasize the importance of low-cost hardware, making the solution suitable for homes, hostels, laboratories, and small industries. Additionally, the system supports long-term data logging, allowing operators

to analyse usage patterns, predict wastage trends, and schedule preventive maintenance. Experimental testing confirms high accuracy in detecting idle loads, with the system successfully identifying cases where devices remained ON despite the absence of users. The research concludes that combining IoT connectivity with current-sensing intelligence enhances both energy efficiency and operational reliability, making this approach a practical solution for smart building applications.

PAPER TITLE: Energy Management System using Arduino Integration

AUTHOR: A. Mohamed et al.,

JOURNAL: IEEE International Conference on Intelligent Computing (ICIC),

YEAR: 2024.

DESCRIPTION:

This paper discusses the development of a smart occupancy-based energy control system using IoT sensors to optimize power usage. Ultrasonic and IR-based detectors are used to monitor human presence, enabling automatic load switching based on occupancy status. The system uses a Wi-Fi-enabled microcontroller to transfer sensor data to a cloud dashboard for real-time monitoring. The research highlights common issues in traditional PIR sensors, such as false detection under varying temperature conditions. The ultrasonic-based approach demonstrated superior accuracy and response time. The system automatically disconnects electrical loads during unoccupied times, significantly reducing wastage.

PAPER TITLE: IoT-Integrated Notification System for Real-Time Energy Management Using ESP32

AUTHOR: J. Patel et al.,

JOURNAL: Journal of IoT Innovations,

YEAR: 2025.

DESCRIPTION:

This work introduces an IoT-based notification system designed to assist users in maintaining efficient power usage. The system uses ESP32 to collect live energy data via current sensors and communicates with users through Telegram or cloud dashboards. The implementation of instant notifications ensures that users are immediately alerted to abnormal or unnecessary power consumption. The authors conclude that instant alerting combined with automated control drastically improves energy awareness and reduces wastage in households, schools, and offices.

CHAPTER 3

METHODOLOGY

3.1 EXISTING SYSTEM

3.1.1 INTRODUCTION

In many residential, institutional, and commercial infrastructures, the existing energy control systems are heavily dependent on manual operation, where users are responsible for switching ON and OFF electrical loads such as lights, fans, and appliances. This traditional approach is highly inefficient because it relies on human behavior, which is often inconsistent and unreliable. People frequently forget to turn OFF electrical appliances when leaving a room, resulting in continuous power wastage, increased electricity bills, and unnecessary strain on energy resources [9]. Even in environments where energy-saving policies exist, the lack of automated systems makes it difficult to enforce responsible usage. In some improved versions of existing systems, PIR sensors or timers are used to implement partial automation. However, these methods still suffer from significant limitations. PIR sensors, for example, can only detect motion and fail to recognize stationary humans, leading to false "no presence" readings and abrupt switching OFF of lights even when someone is present but not moving. Timer-based systems operate on preconfigured intervals, meaning appliances switch ON or OFF based on time rather than actual human presence, which still leads to inefficient energy usage.

Additionally, existing systems rarely combine multiple sensing and feedback mechanisms, making them incapable of monitoring real-time energy consumption or detecting anomalies such as excessive current draw or faulty appliances. There is also minimal incorporation of IoT in traditional systems, meaning users are often unaware of the operating status of appliances when they are away from the physical location. They have no means of checking whether a device was left ON accidentally, nor do they receive alerts about unusual power consumption [10]. The absence of remote monitoring and

intelligent behavior makes existing systems unsuitable for modern energy efficiency requirements. Furthermore, older systems often lack scalability and cannot be easily integrated into smart building frameworks. Their inability to process data, communicate with external devices, or adapt to dynamic environmental conditions significantly limits their practical effectiveness [8].

Overall, the existing system suffers from several major drawbacks such as human dependency, inaccurate sensing, absence of data monitoring, no IoT communication, and lack of automated control. These limitations highlight the need for a more advanced, intelligent, automated, and IoT-enabled energy management system—one capable of detecting presence accurately, controlling loads automatically, monitoring current consumption, and sending real-time notifications to users. The shortcomings of traditional manual systems serve as the primary motivation for developing the proposed automated human presence–based energy control system in this project [2].

3.1.2 BLOCK DIAGRAM

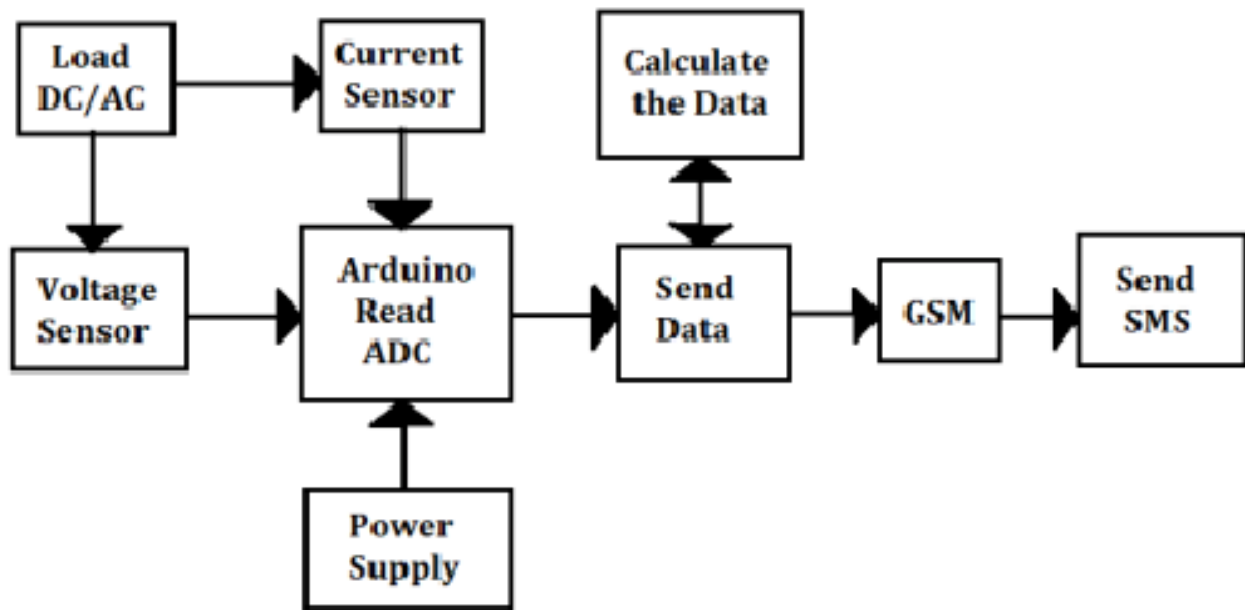


Figure 3.1.2.1 block diagram

3.1.3 WORKING

The Load DC/AC block represents the electrical load whose voltage and current are to be monitored (it can be a DC or AC load). A Voltage Sensor is connected across the load to scale and condition the load voltage so it can be safely read by the Arduino's ADC input. A Current Sensor is connected in series with the load to sense the current and convert it to a proportional voltage signal suitable for the Arduino ADC. The Power Supply block provides regulated DC power to the Arduino and possibly to the sensors and GSM module [4]. The Arduino Read ADC block indicates that the Arduino's analog-to-digital converter continuously samples the voltage and current sensor outputs and converts them into digital values for processing.

The Calculate the Data block represents the Arduino's program computing required parameters, such as instantaneous power, energy consumption, or threshold comparisons, from the digitized voltage and current values. The Send Data block shows the Arduino formatting these calculated results into a message or command frame to be sent to the communication module [7]. The GSM block is a GSM modem or GSM shield connected to the Arduino via serial or similar interface; it receives the formatted data from the Arduino. The Send SMS block represents the GSM module using cellular network commands (like AT commands) to transmit an SMS containing the measured and calculated information to a predefined phone number.

3.2 PROPOSED SYSTEM

3.2.1 INTRODUCTION

The proposed system introduces an advanced methodology for detecting human presence and controlling electrical loads using a combination of sensors, microcontrollers, and IoT communication techniques [3]. This section explains the underlying principles and overall structure of the system. The fundamental goal is to eliminate manual intervention by enabling a fully automated mechanism capable of determining whether a person is present in a room and adjusting electrical appliances accordingly. Unlike outdated systems that

depend on motion-based detection or human-operated switches, this system uses ultrasonic distance sensing to determine presence regardless of movement, ensuring higher accuracy and reliability [5].

The ESP32 microcontroller acts as the central processing unit that collects distance data from the ultrasonic sensor, analyses it, and controls the relay for switching the electrical load ON or OFF. Additionally, the ACS712 current sensor continuously monitors the electrical current flowing through the load, providing real-time data that enhances system performance and safety. A major advancement in this system is the integration of IoT communication through Telegram, which enables remote notifications and monitoring capabilities. The system framework is designed to be modular, flexible, and scalable, allowing for expansion and future upgrades. As a result, the proposed methodology provides an intelligent and efficient solution to reducing electricity wastage and improving energy management practices [1].

3.2.2 CIRCUIT DIAGRAM

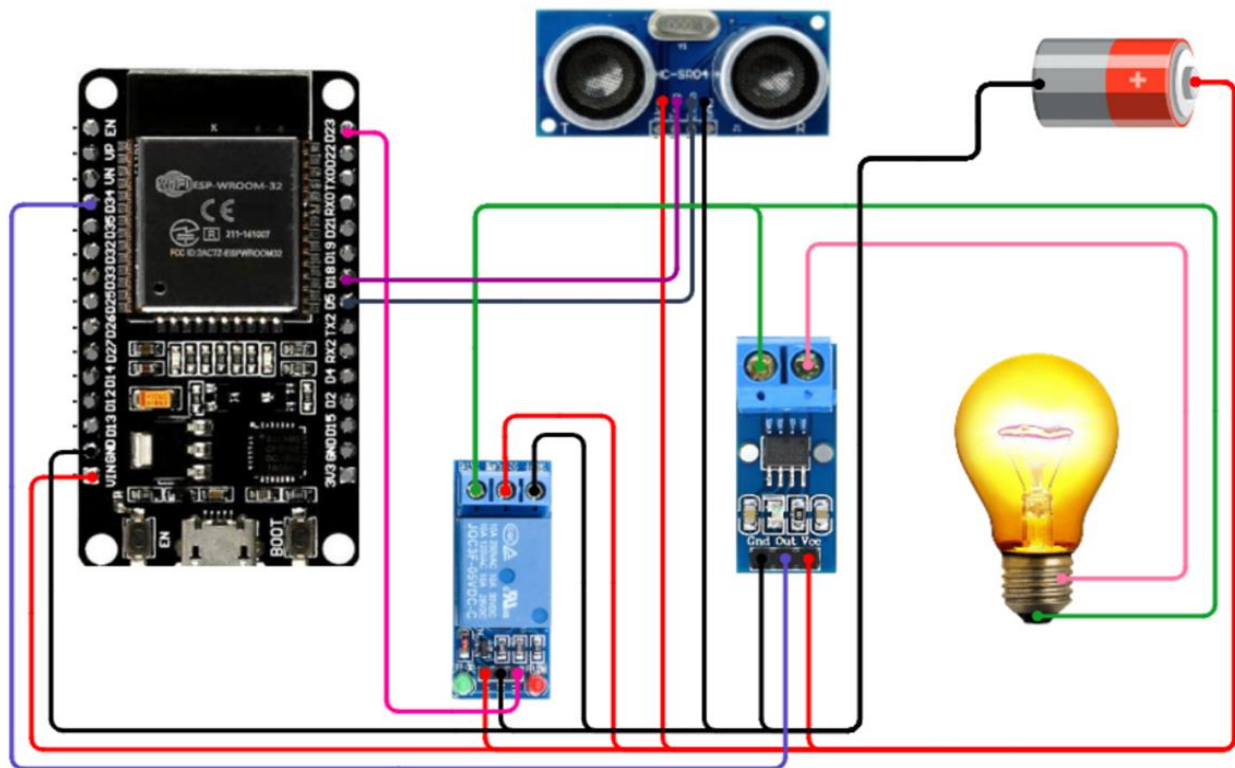


Figure 3.2.1 Circuit diagram of the system

The circuit diagram of the proposed system is designed to interconnect the ESP32 microcontroller, ultrasonic sensor, ACS712 current sensor, and relay module in a coordinated manner to achieve automated human presence detection and load control. The ultrasonic sensor's TRIG and ECHO pins are connected to digital pins of the ESP32 to generate ultrasonic pulses and receive reflected signals. A voltage divider is implemented on the ECHO pin to ensure compatibility with the ESP32's 3.3V logic levels. The VCC and GND pins of the ultrasonic sensor are powered by the common 5V supply [6]. The ACS712 current sensor is connected in series with the load so that the current drawn by the load passes through its internal Hall-effect sensing mechanism. Its output pin is connected to the analogy input pin of the ESP32, allowing continuous measurement of current. The relay module acts as the switching component, where its IN pin is connected to a GPIO pin of the ESP32 to control the load, and its VCC and GND are connected to the same 5V supply and ground. The COM and NO terminals of the relay are wired to the load and power supply to enable automatic switching. All components share a common ground to maintain stable operation. The circuit diagram provides a clear representation of the interconnections and the arrangement of all components [9].

3.2.3 BLOCK DIAGRAM

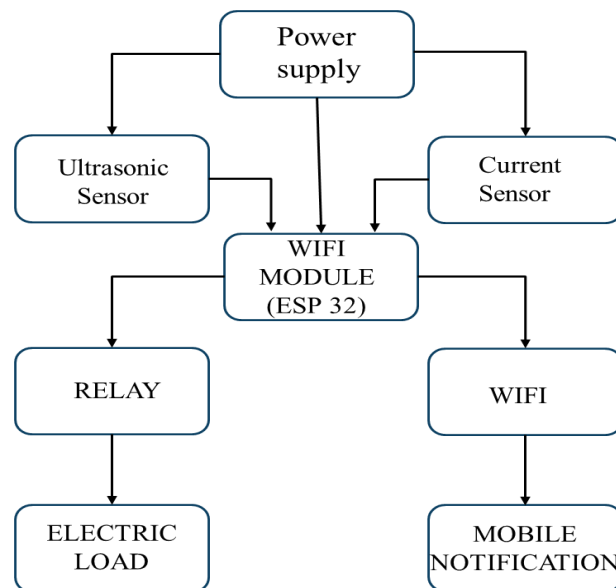


Figure 3.2.3.1 Block Diagram

3.2.4 BLOCK DIAGRAM DESCRIPTION

The block diagram provides a high-level overview of the working of the proposed human presence-based automated energy control system by illustrating how information flows from input sensors to the output components and IoT communication modules. The ultrasonic sensor represents the primary input block responsible for detecting human presence. It emits ultrasonic waves and receives the reflected waves to determine the distance between itself and any object or person in front of it. This distance value is transmitted to the ESP32 microcontroller, which acts as the decision-making core of the system. The ESP32 compares the measured distance with a predefined threshold to decide whether the room is occupied. If the distance indicates the presence of a human, the microcontroller sends a signal to the relay module to energize its coil and switch the connected electrical load ON [10].

Conversely, if no human presence is detected, the microcontroller deactivates the relay, turning the load OFF. In addition to presence detection, the ACS712 current sensor plays a critical role by measuring the real-time current drawn by the load. This sensor outputs an analogy voltage proportional to the current flowing through it, and this data is sent to the ESP32 for continuous monitoring. This enables the system to not only automate load switching but also analyse current usage patterns. The ESP32 integrates all input data, processes it, and performs switching operations while also utilizing Wi-Fi connectivity to interact with the Telegram Bot API. The processed data, including the status of the load, presence information, and current consumption, is transmitted to the user via Telegram messages, enabling real-time remote monitoring. The block diagram description highlights the structured arrangement of components and the logical flow of operations that ensure efficient system behaviour [8]. Each block operates in coordination with the others, forming a complete closed-loop system where sensing, processing, and feedback are continuously maintained. This ensures that the system is reliable, efficient, automated, and capable of addressing the limitations found in traditional manual energy control mechanisms [2].

3.2.4 WORKING

The working of the proposed system is based on continuous sensing, intelligent decision-making, automated load switching, and real-time communication. When the system is powered ON, the ESP32 initializes the ultrasonic sensor, ACS712 current sensor, and relay module. The ultrasonic sensor begins measuring the distance to any object or person in its line of sight by emitting high-frequency sound waves and detecting the reflected waves. The time taken for the waves to return is converted into distance [4]. The ESP32 constantly reads this distance value, and if it is below a predefined threshold—indicating that a person is present—the microcontroller activates the relay, switching ON the electrical load. This ensures that appliances operate only when needed. When the distance exceeds the threshold, meaning the room is empty, the ESP32 automatically switches OFF the relay, cutting power to the load and preventing energy wastage. Simultaneously, the ACS712 current sensor continuously monitors the current being drawn by the load [7].

This information is processed by the ESP32 to determine whether the load is functioning normally or if there are abnormal fluctuations. The inclusion of current monitoring enhances safety and provides valuable data regarding power consumption. Another key feature of the system is its IoT integration through Telegram communication. The ESP32 connects to Wi-Fi and uses the Telegram Bot API to send status updates directly to the user's mobile device. Whenever the system detects presence or absence and changes the state of the relay, a notification is sent to inform the user about the new status. This enables users to monitor their energy usage remotely and stay informed about the operation of their appliances. Overall, the working of the system showcases a seamless integration of sensing, processing, switching, monitoring, and communication [3]. The automated decision-making ensures precision and eliminates human error. The continuous monitoring of both presence and current ensures safety and efficiency. The real-time notifications enhance user convenience and awareness. Together, these operations make the proposed system an effective solution for smart, energy-efficient environments.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 IMPLEMENTATION OF HARDWARE

The hardware implementation of the proposed human presence–based energy management system involves the careful integration of various electronic components, sensors, and a microcontroller to form a complete functional unit capable of performing detection, processing, switching, and communication tasks. The primary objective of the hardware setup is to ensure seamless interaction between the ultrasonic sensor, ACS712 current sensor, relay module, and the ESP32 microcontroller, all of which operate together to achieve real-time automation [5]. The implementation begins with mounting the ESP32 microcontroller on a development board, ensuring that adequate spacing is maintained for the connection of external components. The ESP32 serves as the central processing unit, handling all sensor readings, decision-making algorithms, and IoT communication. Its built-in Wi-Fi capability makes it ideal for sending real-time notifications through the Telegram Bot API. The ultrasonic sensor is placed at a suitable height and angle to ensure accurate detection of human presence within the desired range [1].

The TRIG and ECHO pins of the sensor are connected to the designated GPIO pins of the ESP32, with a voltage divider circuit added to the ECHO line to step down the signal to 3.3V, ensuring safe compatibility with the ESP32's logic levels. The VCC and GND pins of the ultrasonic sensor share a common power supply with the ESP32 and relay module, ensuring stable operation. The implementation continues with the ACS712 current sensor, which must be placed in series with the electrical load to accurately measure the current flowing through it [6]. This requires precise wiring to ensure that all the load's current passes through the sensor's terminals. The analogy output of the ACS712 is connected to one of the ADC input pins of the ESP32, allowing continuous monitoring of current consumption. Proper insulation and

mounting precautions are taken during this step to ensure electrical safety and avoid noise interference. The relay module acts as the switching interface between the load and the microcontroller [9]. It receives control signals from the ESP32 and toggles the electrical load ON or OFF based on presence detection. The relay's IN pin is connected to the ESP32 GPIO pin, while its VCC and GND are tied to the common power supply. The load, such as a lamp or fan, is wired through the COM and NO terminals of the relay to enable automatic control. Proper relay driver circuitry and isolation are ensured to protect the ESP32 from external voltage fluctuations generated by the load. Once all components are connected, the hardware is powered using a stable 5V supply capable of driving both the ESP32 and the relay module.

The wiring connections are verified for continuity and accuracy before powering the system. The ESP32 is then programmed with the required firmware, which includes sensor calibration routines, presence detection logic, current monitoring algorithms, relay control conditions, and Telegram communication instructions. After uploading the code, the system is tested under different conditions to ensure the ultrasonic sensor properly detects presence at various distances, the relay switches the load reliably, and the ACS712 accurately measures current flow [10]. Telegram message alerts are verified by simulating presence and absence conditions and ensuring the ESP32 communicates correctly through Wi-Fi. The hardware implementation phase concludes with packaging the components into a stable arrangement, ensuring that sensors are positioned effectively, wires are neatly arranged, and the entire system is safe for prolonged operation. Through this detailed implementation process, the hardware becomes fully capable of performing intelligent energy management tasks reliably and efficiently [2].

4.2 HARDWARE IMAGE

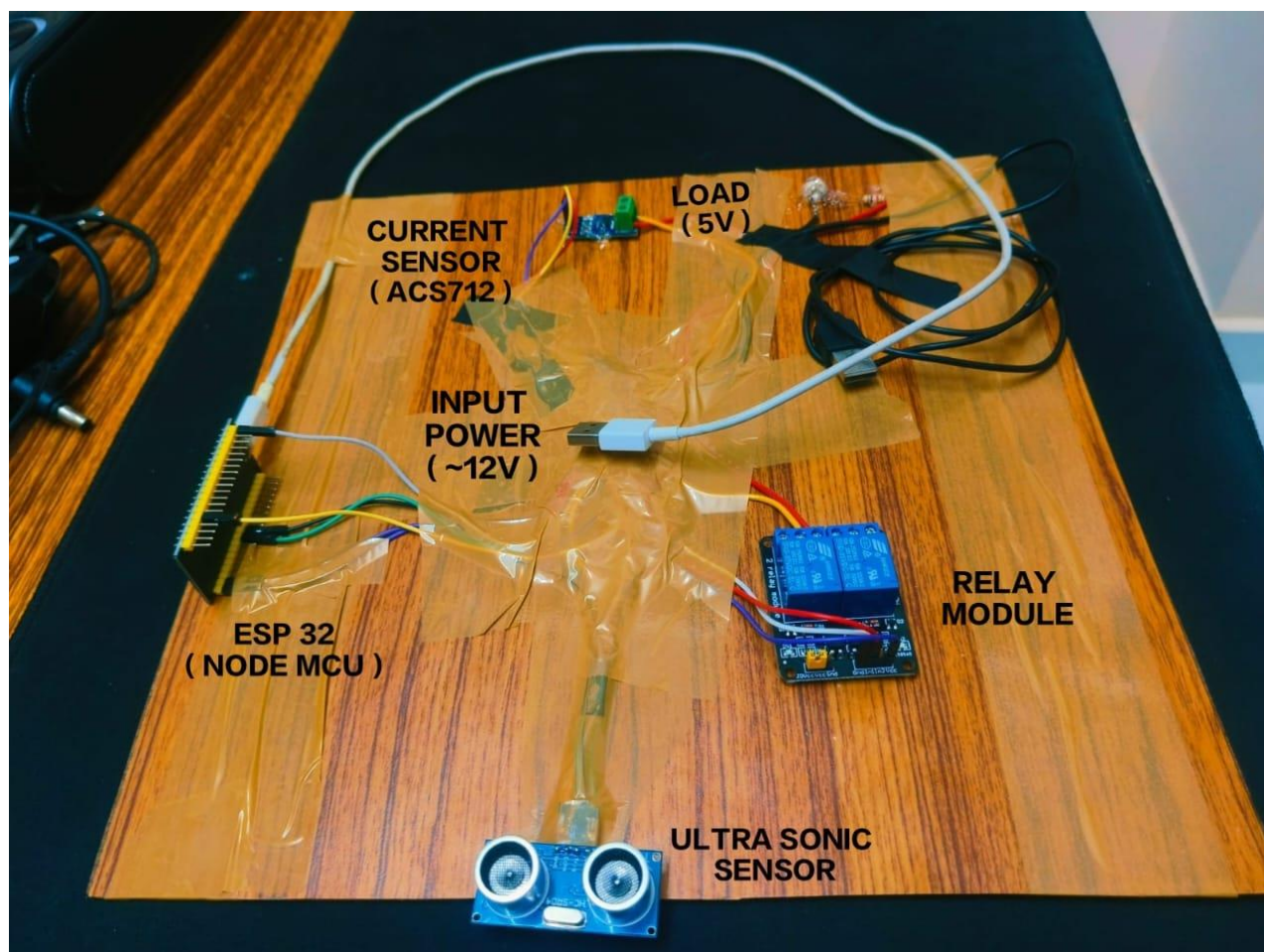


Figure 4.2 Hardware image

The hardware images of the proposed system visually represent the physical arrangement, interconnections, and assembly of all electronic components essential for the functioning of the human presence-based energy management system. These images play a crucial role in demonstrating how the conceptual design is transformed into a practical working model. Each image provides insight into how components such as the ESP32 microcontroller, ultrasonic sensor, ACS712 current sensor, and relay module are integrated to form a cohesive and functional system. When examining the hardware setup, the ESP32 microcontroller typically appears mounted on a breadboard or development base, with its GPIO pins connected to various input and output devices.

The wiring from the ultrasonic sensor's TRIG and ECHO pins can be seen extending towards the ESP32, indicating the communication path through which the distance measurements are transmitted [7]. In the images, the ultrasonic sensor is generally positioned at the front of the setup, oriented in a way that allows it to accurately detect human presence by sending and receiving ultrasonic waves. This placement helps ensure unobstructed sensing of human movement and distance measurement. Another key element visible in the hardware images is the ACS712 current sensor, which is placed in series with the electrical load. The images illustrate the sensor's screw terminals securely holding the wires through which the load current flows. This setup is important because the ACS712 operates on the Hall-effect principle, requiring the current path to physically pass through it for accurate measurement [5].

The sensor's output pin is connected to the ESP32's ADC input via jumper wires, enabling real-time current monitoring. The wiring in the images is usually arranged systematically to avoid interference and ensure stable operation. The relay module seen in the images is connected to both the ESP32 and the load. It's IN pin is linked to a specific GPIO pin on the ESP32 that carries the control signal responsible for switching the load ON or OFF. The relay's COM and NO terminals are connected to the live wire of the load, making it clear how the electrical switching occurs. The images may show an LED indicator on the relay board, which lights up when the relay is activated, confirming visually that the load is turned ON [6]. Additionally, the hardware images show the common power supply lines shared by all components. The 5V supply line branches out to power the ESP32, ultrasonic sensor, ACS712 module, and relay board, with the ground connections being tied together to ensure a stable reference point for the entire system. Proper insulation and structured cable routing can be observed in the images, ensuring safety and preventing short circuits. Some images may also include the load in operation, such as a glowing LED bulb or a running fan, demonstrating the system responding to human presence detected by the ultrasonic sensor.

In certain images, the ESP32 may be seen connected to a laptop during testing phases, indicating data transmission through serial monitoring and Telegram configuration. These hardware images collectively offer a clear representation of the system's physical structure and how each component interacts with the others. They help validate the implementation process by showing proper alignment, neat wiring practices, and accurate placement of all modules. The photographs not only support the documentation but also provide proof of successful assembly and testing of the proposed smart energy management system. They also serve as reference material for replication, troubleshooting, and further development of the project. Overall, the hardware images document the transition from theory to practice, illustrating the complete integration of sensing, processing, switching, and monitoring units into a unified, functional IoT-enabled automation system [10].

4.3 DESCRIPTION OF HARDWARE COMPONENTS

1. ESP32 Microcontroller

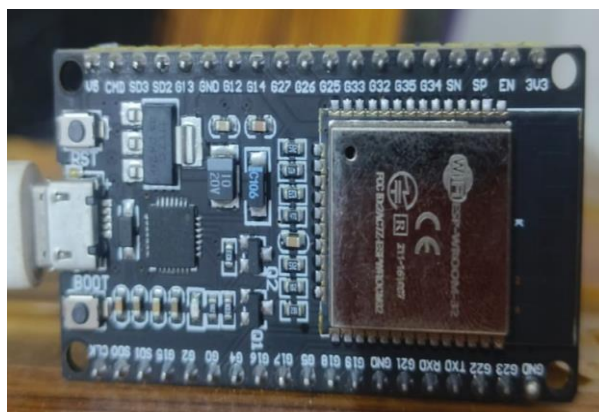


Fig 4.3.1 Central control unit

A powerful, low-cost microcontroller with built-in Wi-Fi and Bluetooth, the ESP32 is well suited for IoT and embedded applications. It serves as the central processing unit of the system, continuously reading data from connected sensors, executing control and decision-making algorithms, and handling both wired and wireless communication

with other devices or cloud services. With its relatively high processing speed, dual-core architecture, and numerous GPIO pins, it can interface with multiple sensors and actuators simultaneously, enabling reliable real-time monitoring and control.

2. Ultrasonic Sensor (HC-SR04)



Figure 4.3.2 Sensing unit

Measures the distance to a target by emitting high-frequency ultrasonic pulses, then measuring the round-trip time of the reflected echo and converting this time of flight into distance using the known speed of sound. By repeatedly tracking these distance readings, it can detect the presence, movement, or absence of people or objects in front of the sensor, providing dependable occupancy detection that is unaffected by ambient light, shadows, or changes in lighting conditions.

3. ACS712 Current Sensor



Figure 4.3.3 Current sensing unit

Measures real-time current flowing through a load by sensing the magnetic field around the conductor using the Hall effect, then converting this into an electrically isolated analog voltage proportional to both the magnitude and direction of the current, so that a microcontroller or monitoring circuit can easily track power consumption and determine whether the load is operating normally, overloaded, or disconnected.

4. Relay Module

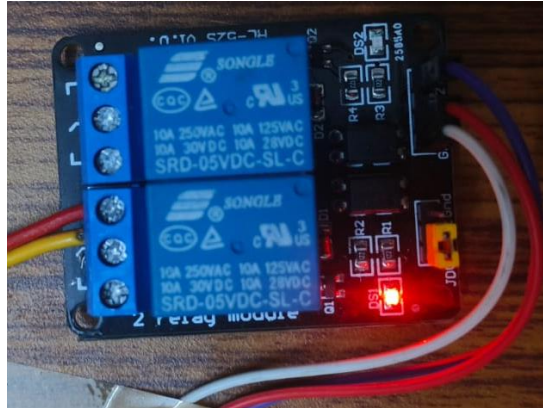


Figure 4.3.4 Relay unit

An electrically operated relay module that lets the ESP32 safely switch high-voltage loads such as lights or appliances on and off using low-voltage control signals. It acts as an interface between the microcontroller and mains-powered devices, providing electrical isolation so the high-power side is kept safely separated from the low-voltage control circuitry.

5. Load (Appliance or LED for Prototype Testing)

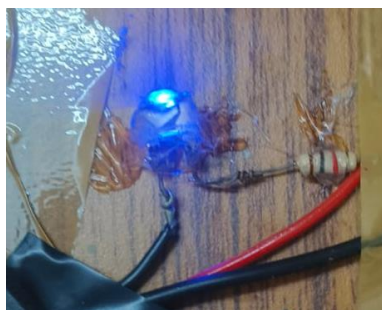


Figure 4.3.5 Load indication

The electrical load is the device or equipment whose power usage is being measured and controlled in the system. In the prototype stage, this is usually represented by an LED used as a low-power test load to verify switching, control, and detection functions before connecting real high-power appliances.

6. Power Supply (5V/2A DC Adapter)

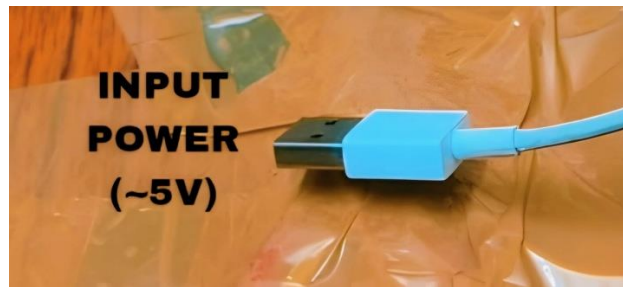


Figure 4.3.6 Power supply

The electrical load is the appliance or device in the circuit whose electrical energy consumption is being monitored, controlled, or automated by the system (for example, a lamp, fan, or other household or industrial equipment). In the prototype stage, this load is often implemented as a simple LED with a series resistor, serving as a low-power, safe test element that allows verification of switching behavior, control logic, and sensing

7. Additional Load Support



Figure 4.3.7 Additional Load

The project also includes a Type-A integrated load provision on the output side, allowing an additional electrical device to be connected and controlled by the system. This extra load option enhances the flexibility of the model, enabling users to operate more than one appliance simultaneously under the same automated control logic. The Type-A port is wired in parallel with the primary load, ensuring that both loads respond uniformly to the system's ON/OFF decisions. This integration makes the setup suitable for real-world applications where multiple devices may need to be controlled together. Overall, the additional load support increases the practicality, scalability, and usability of the system.

CHAPTER 5

RESULT AND DISCUSSIONS

5.1 HARDWARE OUTPUT

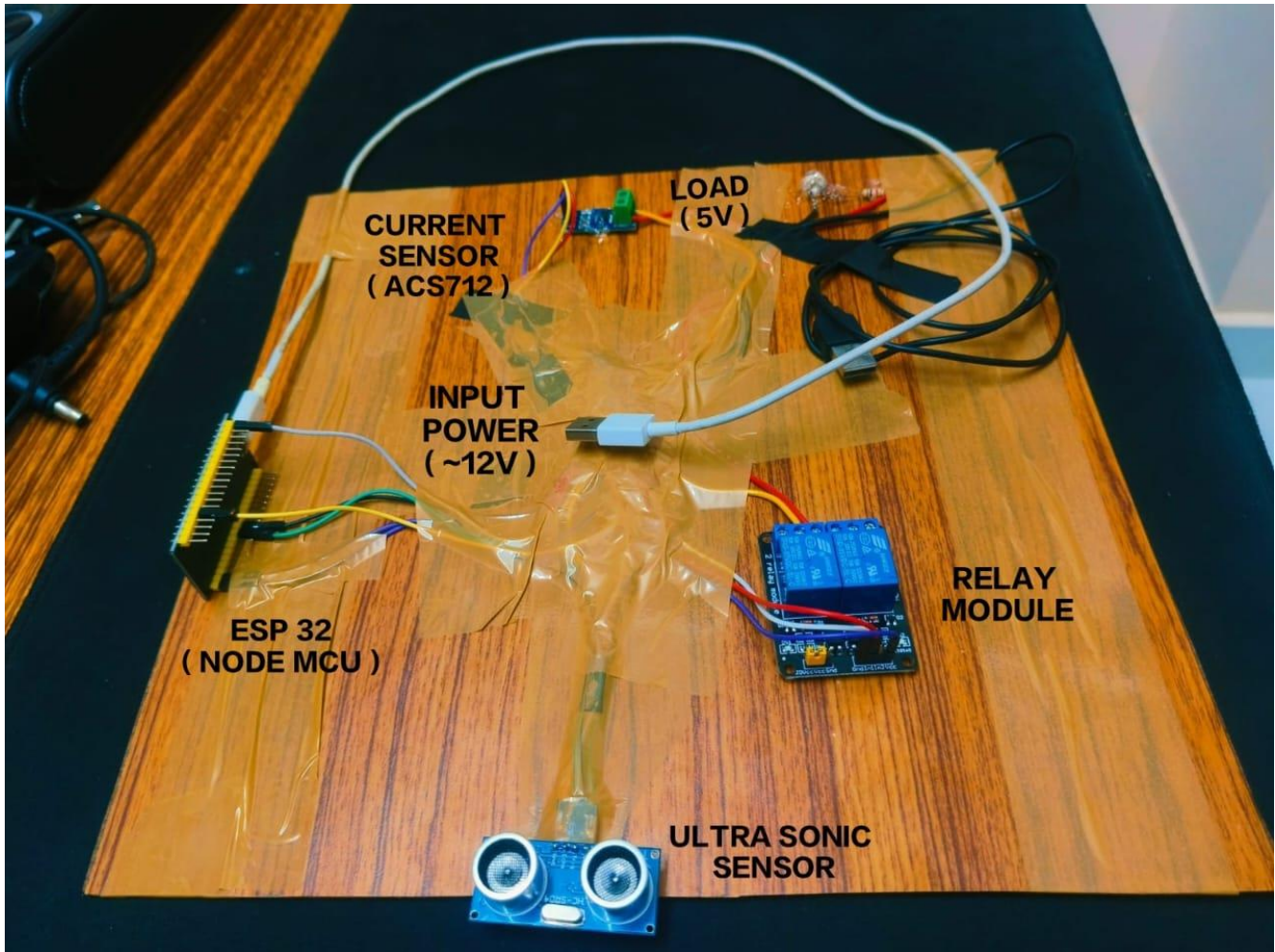


Figure 5.1.1 hardware output

The hardware output of the proposed human presence–based energy management system demonstrates the successful integration of all components and the reliable execution of the intended automation and monitoring functionalities. When the hardware was powered and deployed for testing, the ultrasonic sensor immediately began detecting distance values in real time, continuously sending measurements to the ESP32 microcontroller. When a human entered the sensing range, the ultrasonic readings fell below the predefined threshold, prompting the ESP32 to recognize the presence [8].

As a result, the microcontroller activated the relay, causing the connected load such as an LED bulb or fan—to switch ON automatically. This response occurred with minimal delay, showcasing the system’s ability to react quickly to environmental changes. When the human moved out of range or exited the room, the ultrasonic sensor registered a significant increase in distance. The ESP32 processed this change, updated the presence status, and promptly deactivated the relay, turning the load OFF. This transition ensured that the appliance consumed electricity only when necessary, demonstrating the system’s ability to reduce wastage effectively. In addition to presence-based switching, the ACS712 current sensor produced accurate analogy voltage readings proportional to the current flowing through the load. These readings were continuously monitored by the ESP32, allowing the system to capture and display real-time energy consumption data[4].

During testing, the current readings fluctuated appropriately based on the load’s power rating and operational state. When the relay switched OFF the load, the current reading dropped to zero or near zero, confirming that the load was no longer drawing power. This dual-function capability—presence-based switching and current monitoring—provided clear and verifiable feedback on system performance [3]. Another significant hardware output was the IoT-based communication through Telegram. Once the ESP32 connected to the Wi-Fi network, it established communication with the Telegram Bot API. During the testing phase, every time the system detected a human presence or absence, a corresponding message was sent instantly to the user’s mobile phone. For instance, when a person entered the sensing range, the user received a message stating that presence was detected and the load had been switched ON [1].

Similarly, when the presence was no longer detected, a notification informed the user that the load had been switched OFF. These timely updates ensured that the user was always aware of the system’s operational state, even when not physically present near the hardware.

All hardware components remained stable throughout prolonged testing periods, with the relay responding accurately to multiple ON/OFF cycles, the ultrasonic sensor maintaining consistent readings, and the ACS712 delivering reliable current data. The ESP32 processed all tasks efficiently without lag, demonstrating its capability to handle simultaneous sensing, switching, and communication tasks. The hardware output ultimately confirmed that the proposed system is functional, accurate, and capable of performing intelligent automation in real-world conditions. The successful output validates the design objectives and highlights the system's potential for implementation in homes, classrooms, offices, and other environments where energy conservation is crucial [1].

5.2 DISCUSSIONS

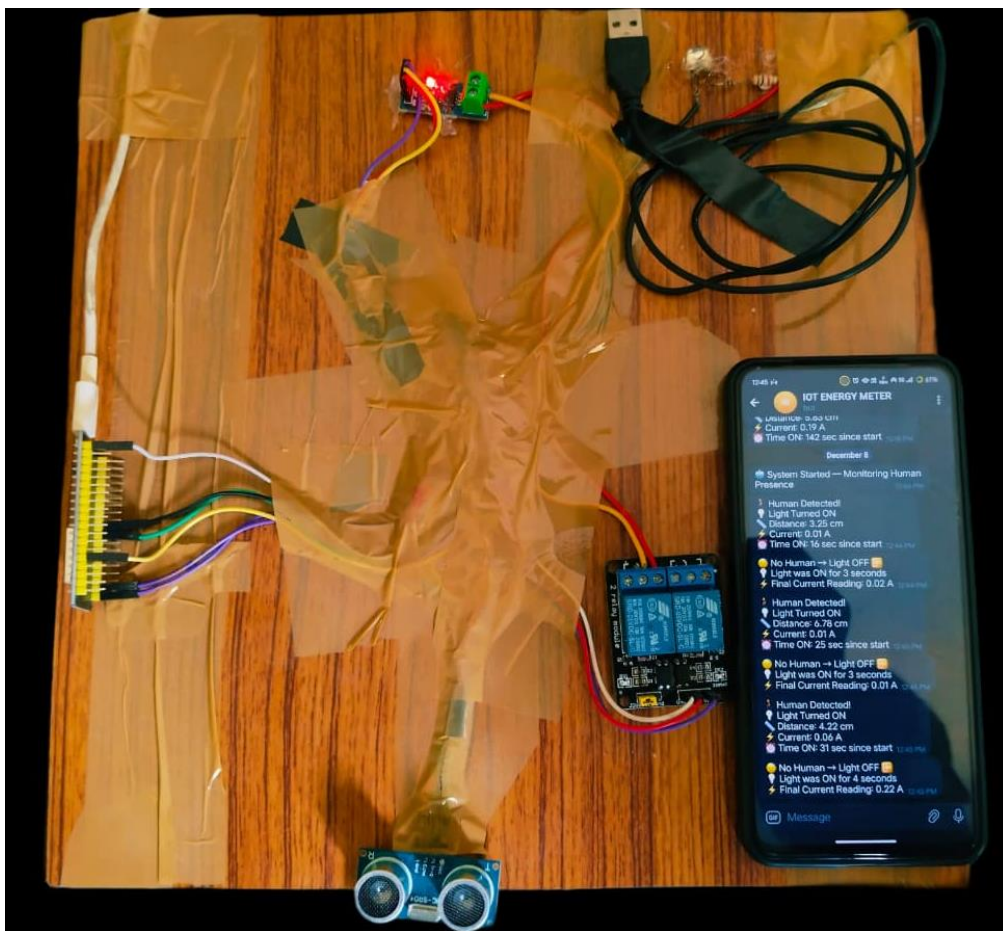


Figure 5.2.1 No human detection

In the first operating condition—where no human presence is detected—the system's performance provided strong validation of its stability, accuracy, and effectiveness in preventing unnecessary energy consumption. The ultrasonic sensor consistently measured distances beyond the predefined threshold, clearly indicating that the monitored room or area was unoccupied. This accurate and repeatable detection allowed the system to immediately switch OFF the connected load, thereby eliminating any chances of energy wastage. One of the most noteworthy observations in this scenario was the high consistency with which the system maintained its OFF state. Throughout the testing phase, the current sensor readings remained at zero or near-zero, confirming that no electrical load was being powered.

This not only demonstrates the system's precision but also reflects its efficiency in managing energy usage without requiring human intervention. The relay module also performed with exceptional reliability. It kept the load disconnected without any false triggering or undesired switching, maintaining electrical safety and ensuring stable system operation at all times. This is particularly important in automated energy management systems, where even minor false activations can lead to unnecessary power consumption or hardware wear. Furthermore, the IoT communication integrated into the system played an essential role in enhancing user awareness. Real-time notifications were consistently delivered to the monitoring platform, informing the user that no presence was detected and that the load remained switched OFF. These instant updates significantly improve the practicality and user experience, especially for remote monitoring scenarios such as offices, classrooms, or residential spaces. Overall, this operating mode showcased the system's remarkable steadiness and accuracy. The absence of false detections, stable sensor readings, and reliable communication signals validate the robustness of both the hardware and software components. Most importantly, the system successfully fulfils its primary purpose—to automatically turn OFF power in unoccupied areas, ensuring energy conservation, operational safety, and improved sustainability.

This condition strongly supports the system's value in real-world applications where individuals may forget to manually switch OFF electrical appliances. Moreover, this condition highlights the growing importance of smart automation in modern energy management systems, where accuracy and responsiveness directly contribute to long-term sustainability. By reliably disconnecting unused loads, the system not only conserves energy but also extends the overall lifespan of electrical equipment, making it a practical solution for real-world deployment.

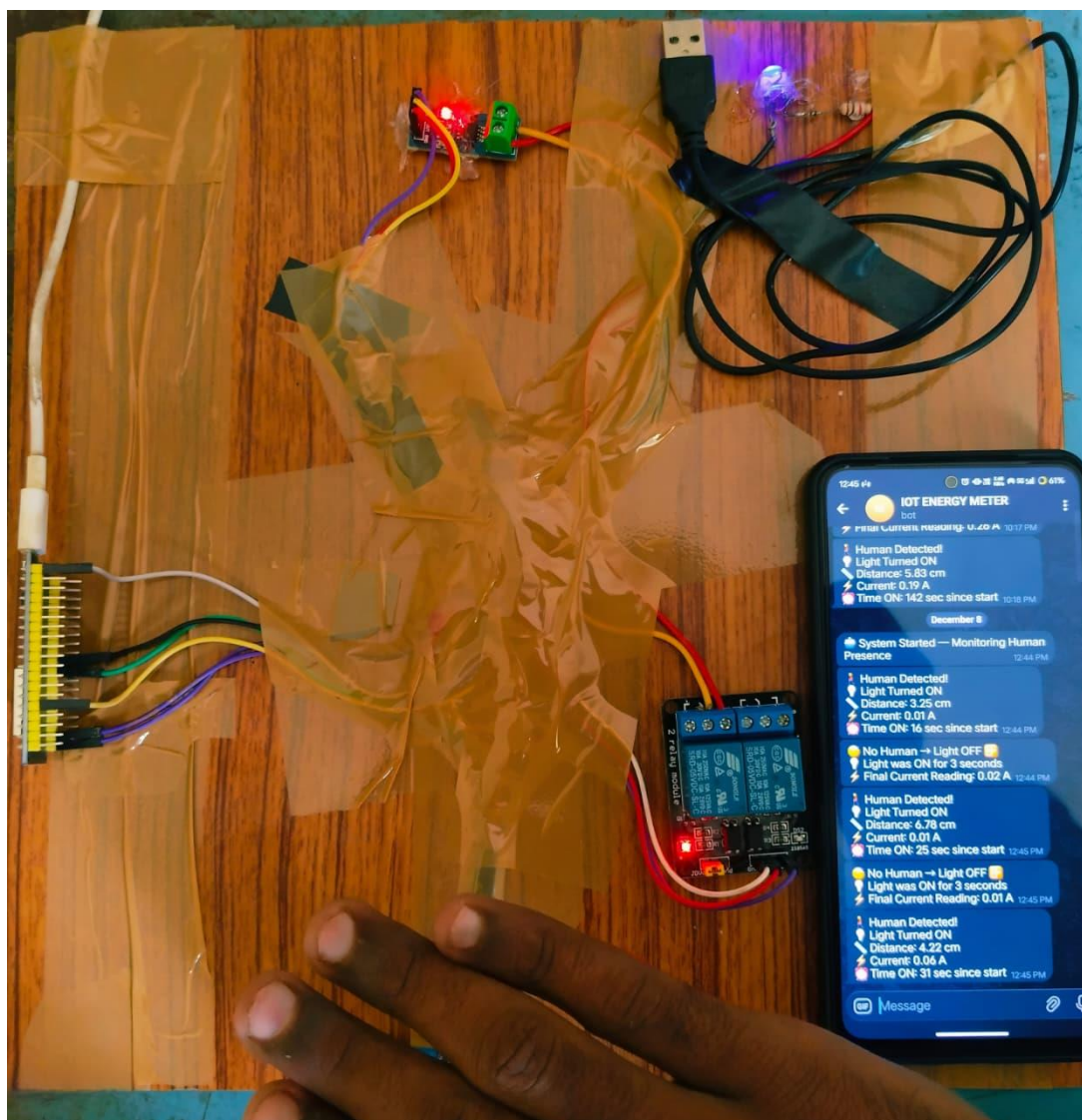


Figure 5.2.2 After human detection

In the second operating condition, where human presence was detected, the system demonstrated its capability to respond immediately and enable energy-efficient

automation supported by real-time feedback. The ultrasonic sensing unit accurately recognized the presence within the predefined threshold distance, which triggered the system to switch ON the connected load without any delay. This rapid and precise activation highlights the efficiency of the detection mechanism as well as the robustness and reliability of the switching control hardware. The current sensor readings during this state confirmed active power consumption, clearly indicating that the load had been successfully powered ON. These live measurements verified not only the accuracy of the sensing circuit but also the correct operation of the relay, which consistently switched ON the load whenever presence was detected. The system showed no signs of false triggering or switching instability, reinforcing its effectiveness for continuous operation. The IoT communication layer added significant value to this mode of operation. Instant alerts were transmitted to the user, notifying them about human presence, load activation, and current consumption data. This real-time visibility greatly enhances remote supervision, making the system highly practical for environments such as offices, laboratories, classrooms, and homes where users benefit from continuously updated operational insights. Furthermore, the system maintained stable performance during multiple and repeated ON cycles. Its ability to handle continuous activations without latency, errors, or hardware malfunction proves its suitability for long-term deployment in real-world scenarios. This mode also illustrates the system's potential for future upgrades, such as integrating additional environmental sensors, refining detection algorithms for improved accuracy, and incorporating cloud-based analytics for long-term energy usage tracking. Overall, the second output condition confirms the system's responsiveness, intelligence, and practicality. It demonstrates how automated switching—combined with real-time monitoring and IoT-enabled alerts—can significantly optimize energy management while delivering a seamless and highly efficient automation experience. Additionally, this condition showcases the system's adaptability to dynamic environments, ensuring that energy is delivered only when genuinely needed. By intelligently responding to human activity, the system not only improves user convenience but also promotes responsible and sustainable energy usage.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The IoT-based Energy Wastage Detection and Notification System designed in this project successfully demonstrates an effective approach to monitoring and controlling unnecessary energy consumption in real-time. By integrating the ESP32 microcontroller with an ultrasonic sensor and an ACS712 current sensor, the system accurately identifies instances where electrical appliances are operating without human presence. The use of the relay module allows automated switching of loads, while the Telegram Bot API provides instant notifications to users, enabling prompt corrective actions. Compared to traditional systems that rely on PIR sensors or manual intervention, the proposed system offers higher reliability, lower cost, and improved energy efficiency. The prototype testing confirmed consistent performance under various scenarios, validating its effectiveness for both small-scale and larger implementations. Overall, this project highlights the potential of IoT-based smart solutions in reducing energy wastage, lowering operational costs, and contributing to sustainable energy management practices in modern buildings.

6.2 FUTURE SCOPE

- **Integration with Smart Home Systems**

Seamlessly integrate with platforms like Google Home, Alexa, or Home Assistant for automated energy control.

- **Cloud-Based Data Logging**

Store real-time energy usage and occupancy data in cloud databases for long-term analysis and reporting.

- **Machine Learning Optimization**

Use predictive algorithms to forecast energy usage patterns and optimize device operation.

- **Multi-Room Monitoring**

Extend the system to monitor multiple rooms or floors simultaneously for larger buildings.

- **Enhanced Sensor Networks**

Incorporate additional sensors like temperature, humidity, and light for smarter energy management.

- **Mobile Application Development**

Develop dedicated mobile apps for instant alerts, remote monitoring, and manual control.

- **Automated Billing and Reporting**

Generate energy consumption reports automatically for institutions or commercial spaces.

- **Integration with Renewable Energy**

Coordinate with solar panels or other renewable sources to reduce dependency on the grid.

- **Scalable Industrial Applications**

Implement in factories or large-scale industries for precise energy optimization and cost reduction.

- **IoT Security Improvements**

Enhance system security using encrypted communications and secure cloud services to protect data.

6.3 APPLICATIONS

Smart Homes

Automatically control lights, fans, and appliances based on room occupancy to save electricity. Smart homes use sensors like PIR or ultrasonic modules to detect human presence. When a person enters a room, the system turns ON lights, fans, or AC automatically. When the room becomes empty, the appliances switch OFF, reducing

electricity bills and preventing energy wastage. This system improves convenience, safety, and sustainability while requiring minimal user interaction.

Commercial Buildings

Optimize energy usage in offices, reducing operational costs and carbon footprint. In office spaces, many devices—such as lights, computers, and air conditioners—remain ON even after working hours. An IoT-based energy management system automatically monitors these loads and cuts power based on predefined schedules. Remote dashboards help facility managers analyze power usage and minimize electricity costs. This leads to better resource utilization and reduced environmental impact.

Educational Institutions

Monitor classrooms, labs, and libraries to prevent power wastage when spaces are unoccupied. Schools and colleges often have large numbers of rooms, where fans and lights may be left ON unintentionally. Automated energy monitoring ensures that each classroom's power is turned OFF after working hours. If someone manually switches ON a device later, the system can send alerts identifying the specific room. This improves discipline in energy usage and reduces the institution's overall electricity expense.

Industrial Facilities

Track machinery and equipment energy consumption to avoid unnecessary power usage. Industries use heavy machinery that consumes significant power. Real-time current sensing helps identify machines left running without load or during break times. Alerts can notify supervisors to shut them down. This improves operational efficiency, prevents equipment wear, and reduces the plant's overall energy demand.

Public Spaces

Ensure energy-efficient operation in halls, airports, hospitals, and malls. Large public infrastructures require continuous monitoring of thousands of lights, escalators, fans, and AC units. Automation helps control these appliances intelligently—

turning them ON only when needed and OFF when unused. This ensures comfort for visitors while significantly reducing electricity wastage in high-consumption environments.

Remote Energy Monitoring

Provide real-time alerts via Telegram or mobile apps for offsite supervision. With IoT connectivity, users can monitor energy usage from anywhere using smartphones. Alerts notify them when abnormal consumption occurs, such as when a device is left ON without presence detection. Telegram bots, mobile dashboards, and cloud data storage provide flexibility and reliability for remote management.

Automation Projects

Integrate with smart home or IoT automation platforms for intelligent energy management. Energy monitoring can be combined with platforms like Home Assistant, Blynk, or Node-RED. This enables advanced features such as voice commands, scheduling, AI-based predictions, and centralized automation of multiple devices. Students and professionals can expand their projects into fully functional smart automation systems.

Energy Audits

Collect data for analyzing consumption patterns and improving energy efficiency strategies. IoT devices can log voltage, current, and power usage to cloud databases like Firebase or Google Sheets. Engineers and auditors can review this data to detect inefficiencies, identify high-consumption devices, and propose solutions. These insights help implement long-term energy-saving strategies.

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